

Kinshuk Gupta

[Leetcode](#) [GFG](#) [Codechef](#) [Codeforces](#) [Github](#) [LinkedIn](#) [+91 9479003023](#)
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Education

Indian Institute Of Technology (IIT), Bhilai

Aug 2023 – May 2027

BTech in Data Science & Artificial Intelligence

Kendriya Vidyalaya No. 1, Raipur (CBSE)

Class XII (2022): **94.6%** | Class X (2020): **94%**

Experience

AI Product Development Intern — ARPN Consulting Services (STPI, Bhilai)

May 2026 – Jul 2026

- **Architected** a scalable enterprise backend for a billing application using **Python (Flask)** and **SQL databases**, streamlining complex data pipelines for Customs claims.
- **Developed** and evaluated backend integration workflows in an agile environment, enforcing code quality through continuous iteration and rigorous API testing via **Postman**.
- **Deployed** containerized machine learning architectures to a staging cloud server using GitOps, collaborating cross-functionally to ensure system reliability and low-latency access.

Achievements & Competitive Programming

- **The Integral Cup Season 2 (Jane Street & Optiver):** Secured **All India Rank 110** out of 7,500+ elite candidates in Round 1. Bypassed intense time constraints and complex mathematical roadblocks, qualify for the exclusive Round 2.
- **Algorithmic Mastery:** Solved **2500+** algorithmic problems across Codeforces, CodeChef, LeetCode, and GeeksforGeeks.
- **IICPC 2025:** Secured **Rank 1102** in the Indian ICPC (International Collegiate Programming Contest) 2025.
- **GeeksforGeeks Consistency:** Awarded exclusive GFG merchandise (T-Shirt) for accumulating **800+ coins** through rigorous daily problem-solving (POTDs) and contest participation.
- **Hackathons:** Secured **3rd position** at The Forge Hackathon (Meraz 6.0) for architecting StudySphere, a data-driven productivity analytics app.

Technical Skills

Languages: Python, C++, C, SQL, PySpark, Bash Scripting, JavaScript, TypeScript, HTML/CSS

Web, Cloud & DevOps: AWS (Amazon Web Services), Docker, Flask, FastAPI, Node.js, RESTful APIs, Linux, Git

Libraries / Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, HuggingFace, LangChain, LlamaIndex

Databases & Developer Tools: MySQL, MongoDB, Postman, CUDA, TensorRT, Qlik

Core Competencies: Data Structures & Algorithms, Object-Oriented Programming (OOP) in C++, Machine Learning & AI, Database Management Systems, Operating Systems, Computer Architecture, Computer Networks, Distributed Computing, MCP (Model Context Protocol), AI Agents, Computer Vision, Image Processing, Natural Language Processing (NLP), Retrieval Augmented Generation (RAG), Agile Methodologies, Unit/Automation Testing, Quantum Computing

Projects

Inter IIT Tech Meet 14.0 — GeoNLI (Problem Statement by ISRO SAC)

[Github](#)

Python, FastAPI, scikit-learn, Multimodal Deep Learning, YOLO-OB

- Built **Vision42**, a vision-language system for satellite imagery supporting image captioning, VQA, and text-driven object grounding.
- Designed a hybrid pipeline combining OBB detection with multimodal fusion; systematically evaluated and validated models on GPU infrastructure, securing **6th rank**.

DGCNN on Edge Devices – GNN for Jetson Nano

[Github](#)

Python, PyTorch, Edge AI, CUDA, Hardware Optimization

- Architected an edge-optimized DGCNN for 3D point cloud classification on ModelNet10, achieving **93.4% accuracy** with an **18ms latency** (118K parameters).
- Engineered a Progressive K-Reduction pipeline in custom EdgeConv layers, slashing total KNN edge scatter operations by **37.5%** without sacrificing spatial context.
- Bypassed Maxwell GPU constraints via a custom $O(n^2 \cdot D)$ torch.cdist graph builder and Static Graph Reuse, eliminating **75%** of KNN latency.

Inter IIT Tech Meet 13.0 — Lunar Surface Mapping (Problem Statement by ISRO)

[Github](#)

Python, NumPy, SciPy, Astropy, PyTorch, QGIS

- Worked in a highly **collaborative** team environment to engineer a 4-stage pipeline processing Chandrayaan-2 CLASS XRF Level-1 data to generate high-resolution lunar elemental composition maps.
- Implemented rigorous signal processing techniques, including moving average noise filtration and Gaussian peak fitting, to extract quantitative elemental abundances.
- Accelerated processing time by 40% using PyTorch parallelization, directly impacting project metrics and securing **11th rank**.

Brain Functional Connectivity (fMRI)

[Github](#)

Python, FSL, Nilearn, PyTorch, NetworkX, scikit-learn, MONAI

- **Architected** an fMRI pipeline using **FSL** and **Nilearn** to preprocess 40 ADNI subjects across 12 steps, extracting 1,128-dimensional FC features and using Mixup augmentation to scale the cohort $8\times$.
- **Engineered** a classification framework benchmarking L1-SVMs, ensembles, and **3D CNNs with Grad-CAM**, achieving **97.5% LOOCV accuracy** ($F1 = 0.976$) on the original cohort and **99.4%** on augmented data.
- **Developed** a graph-theoretic resilience analysis via **NetworkX** to simulate hub attacks, revealing paradoxical resilience in MCI networks as computational evidence of compensatory cortical reorganization.